Bonus: Data Quality Monitoring Dashboard

Objective

As part of the unified payment platform initiative, I designed a lightweight Data Quality Monitoring Dashboard to provide visibility into the health of payment data flowing from the Cards, Transfers, and Bill Payments domains.

The goal is to identify data quality issues early, improve trust in the canonical payment model, and provide product squads with actionable insights before issues impact downstream analytics, finance reporting, or customer operations.

Why a Data Quality Dashboard is Needed

In a multi-product banking environment, different squads own different payment systems and release changes independently.

Without centralized monitoring, common issues can go unnoticed:

- Missing customer identifiers
- Invalid payment amounts
- Unexpected status values
- Schema drift
- Delayed data delivery
- Duplicate payment events

A shared dashboard provides a single source of truth for payment data quality across all squads.

Dashboard Design

The dashboard is organized into four sections:

1. Pipeline Health

Purpose:

Monitor whether data pipelines are operating successfully.

Metrics:

Metric	Description
Pipeline Run Status	Success / Failed

Metric	Description
Processing Time	Runtime per execution
Records Processed	Total records ingested
Failed Records	Validation failures
Last Successful Run	Most recent successful execution

Example:

Pipeline Status: SUCCESS

Records Processed: 102,450

Failed Records: 37

Failure Rate: 0.036%

2. Data Quality Metrics

Purpose:

Measure quality of incoming payment events.

Metrics:

Completeness

Percentage of records containing required fields.

Examples:

- customer_id populated
- amount populated
- currency populated

Target:

99%

Validity

Percentage of records passing validation rules.

Examples:

- amount > 0
- valid currency code

- approved status value

Target:

99%

Uniqueness

Percentage of unique event IDs.

Target:

100%

Consistency

Percentage of records matching canonical standards.

Examples:

- standardized status values
- approved payment types

Target:

99%

3. Squad-Level Quality View

Purpose:

Create accountability and transparency across squads.

Metrics by Squad:

- Cards
- Transfers
- Bill Payments

Displayed Metrics:

Squad	Records	Validation Pass Rate	Duplicate Rate
Cards	45,000	99.8%	0.01%
Transfers	38,000	99.5%	0.02%
Bill Payments	19,000	99.9%	0.00%

This allows teams to quickly identify areas requiring attention.

4. Data Freshness Monitoring

Purpose:

Ensure downstream teams receive data on time.

Metrics:

Metric	**Description**
Latest Event Timestamp	Most recent payment event
Processing Delay	Time from ingestion to storage
Freshness SLA	Expected update frequency

Target:

Data available within 15 minutes of ingestion.

Alerts are generated when freshness thresholds are exceeded.

Alerting Strategy

The dashboard is paired with automated alerts.

Alert Conditions:

1. Validation failure rate exceeds 1%
2. Duplicate rate exceeds 0.5%
3. Missing required fields exceed 1%
4. No data received within SLA window
5. Unexpected schema changes detected

Severity Levels:

- Info
- Warning
- Critical

Critical alerts require immediate investigation by the owning squad.

Ownership Model

The dashboard follows a shared ownership approach.

Data Engineering Team:

- Dashboard maintenance
- Validation framework
- Contract enforcement

Product Squads:

- Source data quality
- Issue remediation
- Schema compliance

This model ensures accountability remains with the teams producing the data.

Success Metrics

The effectiveness of the dashboard will be measured through:

1. Validation Pass Rate

Target: >99%

2. Data Freshness Compliance

Target: >95%

3. Duplicate Event Reduction

Target: <0.1%

4. Mean Time to Resolution (MTTR)

Target: <24 hours

5. Canonical Schema Adoption

Target: 100% payment coverage within 90 days

Production Implementation

For the assignment, a simple dashboard can be implemented using:

- Streamlit
- Python
- Pandas

- Parquet output files

For production-scale deployment (100K+ transactions/day), I would integrate:

- Airflow for orchestration
- Great Expectations for data quality validation
- Cloud monitoring dashboards
- Automated alerting integrations
- Historical trend reporting

Expected Business Impact

The Data Quality Dashboard supports the broader goal of creating a reusable payment data platform across Mal's product ecosystem.

Expected outcomes include:

- Increased trust in payment analytics
- Faster issue detection
- Reduced duplicated validation logic
- Improved collaboration across squads
- Easier onboarding of future payment products

By providing a shared view of payment data quality, the dashboard helps ensure that the canonical payment model remains reliable, scalable, and valuable for both operational and analytical use cases.